

Gene Pulser® Electroprotocols

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type Mammalian, adherent

Species Used Rat, fibroblasts

Molecules Electroporated DNA, retroviral vector, pN2, 9 kB, linearized at *Hind* III site.

Before the Pulse

Cell Growth Medium DMEM, 10% Fetal Calf Serum (GIBCO/BRL, Sigma)

Growth Phase at Harvest 70% confluent

Pre-pulse Incubation 2.0 min

Wash Solution None

The Pulse

Instruments Used Gene Pulser® apparatus & Capacitance

Electroporation Temperature 25 °C

Electroporation Medium* DMEM, 10% Fetal Calf Serum

Cuvette Gap 0.4 cm

Cell Density 10 (7) cells / ml

Voltage 0.25 kV

Volume of Cells 250 µl

Field Strength 0.625 kV/cm

DNA Concentration 0.4 µg / µl

Capacitor 500 µF

DNA Resuspension Buffer water

Resistor (Pulse Controller) Ω none

Volume of DNA 5 µl

Time Constant 18.0 msec

After the Pulse

Outgrowth Medium DMEM, 10% Fetal Calf Serum

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

Outgrowth Temperature 37 °C

Length of Incubation 10 days

Selection Method or Assay Used G418 resistance, 500 µl / ml

Electroporation Efficiency 1.21 x 10 (3) transfectants / µg DNA

Per Cent Survival Not recorded

Name of Submitter Marc D. Corina, Graduate Student

Institution Address Baylor College of Medicine
Cell Biology
One Baylor Plaza
Houston, Texas 77030

Survey Number

148